

$^{16}\text{O}(^{24}\text{Mg}, \alpha \text{p} \gamma)$ 2004Ek01, 2005Ek01

2004Ek01, 2005Ek01: E=60 MeV ^{24}Mg beam was produced at the Legnaro National Laboratory. Target was a 0.5 mg/cm² enriched ^{40}Ca target with a 7 mg/cm² tantalum backing and with oxygen inside. γ rays were detected with the GASP array of Ge detectors and charged particles were detected with the ISIS array of 40 $\Delta\text{E-E}$ Si telescopes. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma(\theta)$, $\gamma(\text{particle})$ -coin. Deduced levels, J, π from measured asymmetry ratios $R(\gamma(\theta))$.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$
0.0	$3/2^+$
1762.95 17	$5/2^+$
2645.81 18	$7/2^+$
3163.43 19	$7/2^-$
4348.29 24	$9/2^-$
5407.64 27	$11/2^-$
6088.05 29	$13/2^-$

[†] From a least-squares fit to γ -ray energies.

[‡] As given in 2004Ek01 based on measured $\gamma(\theta)$ asymmetry and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

 $\gamma(^{35}\text{Cl})$

$R(\gamma(\theta))$ =Asymmetry ratio measured at 35° and 81° with respect to the beam axis. Value of ≈ 1.2 for stretched quadrupole and ≈ 0.7 for stretched dipole ($\Delta J=1$).

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	Comments
517.7 1	14.3 8	3163.43	$7/2^-$	2645.81	$7/2^+$		
680.4 1	39.1 12	6088.05	$13/2^-$	5407.64	$11/2^-$	D	$R(\gamma(\theta))=0.65$ 3.
882.9 1	4.9 4	2645.81	$7/2^+$	1762.95	$5/2^+$		
1059.3 2	9.1 9	5407.64	$11/2^-$	4348.29	$9/2^-$		
1185.0 2	32.9 10	4348.29	$9/2^-$	3163.43	$7/2^-$		
1702.0 3	13.0 8	4348.29	$9/2^-$	2645.81	$7/2^+$		
1763.1 2	22.9 7	1762.95	$5/2^+$	0.0	$3/2^+$		$R(\gamma(\theta))=1.43$ 7 indicating Q seems inconsistent with $\Delta J=1$ from level scheme.
2244.2 3	53.9 17	5407.64	$11/2^-$	3163.43	$7/2^-$	Q	$R(\gamma(\theta))=1.24$ 6.
2645.5 4	32.3 10	2645.81	$7/2^+$	0.0	$3/2^+$	Q	$R(\gamma(\theta))=1.21$ 6.
3162.7 4	100 3	3163.43	$7/2^-$	0.0	$3/2^+$	Q	$R(\gamma(\theta))=1.43$ 7.

[†] From 2004Ek01.

[‡] Deduced from $\gamma(\theta)$ asymmetry in 2004Ek01 by the evaluators.

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Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
- $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
- $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

